



Does exposure to biodiversity risk drive firms' digital transformation?

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ABSTRACT

This paper investigates whether biodiversity risk drives firms' digital transformation. We find that such risk positively influences digital efforts, especially among firms facing higher natural disaster exposure, policy uncertainty, and risk perception. The effect operates through disrupted supply chains, sustainability constraints, and incentives for green innovation, positioning digital transformation as a key strategic response to biodiversity challenges.

1. Introduction

Biodiversity—the diversity of genes, species, and ecosystems—is crucial for human prosperity and significantly influences firm value and decision-making (Kalhor and Kyaw, 2024). Previous studies indicate biodiversity risks adversely impact corporate investment efficiency (Yin et al., 2023), financial risk (Liang et al., 2024), capital structure (Ahmad and Karpuz, 2024), and market performance (El Ouadghiri et al., 2025; Garel et al., 2024; Ma et al., 2024; Bach et al., 2025). Yet, effective strategies to manage biodiversity risks remain underexplored.

Generally, digital transformation is viewed as a critical strategy for mitigating risks (Bianchi and Brière, 2024) and enhancing resource allocation efficiency (Cao et al., 2023; Hvide et al., 2024). However, as biodiversity risks escalate, firms face a trade-off between rising transformation costs and uncertainties (Daminato et al., 2024) and the enhanced resilience and growth opportunities digitalization provides (Cong et al., 2024; Gaspar et al., 2024). Our study thus explores digital transformation's role in managing biodiversity risks and investigates its underlying mechanisms.

We employ fixed-effects regressions to examine the relationship between biodiversity risk and firms' digital transformation. Results show a significant positive association, particularly among firms exposed to higher natural disaster risks, increased climate policy uncertainty, and elevated risk perception. Further, we disaggregate digital transformation into Data, Integration of Informatization and Industrialization, Computing Power, and Algorithms, we find firms prioritize these dimensions in this respective order. Our analysis reveals that biodiversity risk primarily fosters digital transformation by increasing supply chain disruption risk, constraining sustainable development, and promoting incentives for green innovation.

This study contributes to the existing literature in three important ways. First, although previous studies have extensively documented the adverse effects of biodiversity risk on corporate investment efficiency (Yin et al., 2023), financial risk (Adamolekun, 2024; Liang et al., 2024), capital structure (Ahmad and Karpuz, 2024), and market performance (Garel et al., 2024; Bach et al., 2025), strategic responses by firms to biodiversity risk remain underexplored. Our study is the first to propose digital transformation as a

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pivotal strategic response to biodiversity risk. We systematically develop a progressive adaptation framework, which advances from data sensing through system integration to intelligent decision-making, enriching the theoretical understanding of the “risk–strategy” mechanism and providing new empirical insights into firms’ adaptive behaviors under biodiversity loss pressures. Second, we innovatively uncover the underlying mechanisms driving firms’ digital transformation in response to biodiversity risk, specifically identifying three critical pathways: supply chain disruption risk, constraints on sustainable development capacity, and incentives for green innovation. This mechanism-focused perspective expands the theoretical frontier concerning how environmental uncertainty reshapes corporate digital strategies and offers robust empirical evidence for the multidimensional channels through which biodiversity risk influences digital transformation. Third, methodologically, we utilize Latent Dirichlet Allocation (LDA) to quantify and distinguish different digital transformation pathways, thereby clearly delineating firms’ strategic priorities in response to biodiversity risks.

2. Hypothesis development

2.1. Biodiversity risk and digital transformation

Firms frequently invest in dynamic capabilities to enhance their adaptability and resilience in response to external uncertainties (Teece et al., 1997). As an emerging form of environmental uncertainty, biodiversity risk increasingly poses challenges to corporate operational stability and risk management capabilities (Liang et al., 2024). Particularly in resource-intensive industries such as food, agriculture, and pharmaceuticals, biodiversity loss can significantly disrupt supply chain stability and hinder access to essential raw materials (Li et al., 2025). Concurrently, international agreements like the Kunming-Montreal Global Biodiversity Framework have prompted governments and regulatory bodies to strengthen oversight of corporate ecological impacts, including implementing mandatory biodiversity information disclosures and due diligence requirements. Additionally, external stakeholders such as ESG rating agencies and green investors increasingly demand transparency regarding firms’ biodiversity-related risks and mitigation strategies. Notably, heightened corporate awareness of biodiversity risks not only reflects proactive risk management but also creates emerging market opportunities in ecological restoration and green technologies (Tian and Chen, 2024).

In this context, digital technologies have become essential tools for firms aiming to monitor ecological risks in real-time, optimize resource allocation, and sustain competitive advantage. Specifically, digital tools significantly enhance firms’ capacity to identify risks and forecast scenarios, particularly in volatile ecological environments (Gaspar et al., 2024; Hong et al., 2024; Gambacorta et al., 2023).¹ Consequently, biodiversity risk is increasingly becoming a critical driver of digital transformation, enabling firms to bolster organizational resilience and strategic flexibility. Nevertheless, firms must carefully balance the rising costs associated with digital transformation (Daminato et al., 2024) against the benefits of improved resilience to external shocks (Cong et al., 2024). Accordingly, we propose the following hypothesis:

H1. *Corporate exposure to biodiversity risk is positively associated with digital transformation.*

2.2. Mechanism of supply chain disruption risk

The degradation of biodiversity directly erodes natural capital, leading to shortages in raw materials and increasing production volatility, particularly within resource-dependent sectors (Li et al., 2025). Firms operating in fragile ecosystems are more susceptible to biodiversity-induced disruptions, such as pollinator declines, water scarcity, and deforestation (Kalhor and Kyaw, 2024). These disruptions propagate along supply chains, amplifying firms’ vulnerability to systemic shocks (Osadchiy et al., 2016). Therefore, we propose:

H2. *Corporate exposure to biodiversity risk is positively associated with supply chain disruption risk.*

2.3. Mechanism of sustainable development

Sustainable development heavily relies on the availability and renewability of ecosystem services. Biodiversity loss disrupts ecological balance, constraining firms’ capacity to fulfill environmental objectives and diminishing their long-term growth potential (Giglio et al., 2023). Once biodiversity thresholds are surpassed, ecological constraints intensify, threatening the viability and sustainability of business models (Wagner, 2023). Meanwhile, Cong et al. (2024) emphasize that digitalization significantly enhances organizational resilience by stabilizing cash flows, mitigating environmental shocks, and supporting sustainable growth. Hence, we propose:

H3. *Corporate exposure to biodiversity risk is negatively associated with sustainable development.*

¹ Specifically, Nestlé employs satellite remote sensing and AI algorithms to develop a digital supply chain system, enabling real-time monitoring of cocoa and palm oil cultivation impacts on biodiversity. Google introduced “Google Earth Engine,” assisting governments and enterprises in remotely monitoring biodiversity metrics to integrate digital and sustainable business strategies. Alibaba’s “Ant Forest” initiative leverages digital incentives and integrates remote sensing, AI, and blockchain technologies for real-time monitoring of vegetation restoration and biodiversity changes. JD.com integrates AI, blockchain, and IoT in its smart supply chain, creating a platform for biodiversity risk management that ensures ecological traceability from production to consumption.

2.4. Mechanism of green innovation

In response to mounting ecological pressures and institutional demands, firms commonly adopt legitimacy-oriented strategies, such as green innovation. Biodiversity risk, intensified by stakeholder scrutiny and regulatory pressures, increasingly motivates firms to invest in environmentally friendly technologies to ensure compliance with sustainability standards (Tian and Chen, 2024). Firms that proactively adopt sustainable practices are better positioned to respond effectively to regulatory and stakeholder demands, maintain operational stability, and adhere to biodiversity and environmental standards (Treepongkaruna, 2024). Consequently, we propose:

H4. *Corporate exposure to biodiversity risk is positively associated with green innovation.*

Our research framework is illustrated in Fig. 1.

3. Research design

3.1. Sample and data

Our sample consists of Chinese-listed firms from 2007 to 2023. We exclude firms in the financial industry, special treatment firms, and those with abnormal or missing data. The final dataset includes 33,983 firm-year observations. The financial data is sourced from the CSMAR database.

To quantify biodiversity risk, we use the indicator developed by He et al. (2024). Biodiversity risk is measured as the proportion of characters related to biodiversity in the annual report to the total number of characters in the report (*Bio-Risk*) (<https://sites.google.com/view/brianmlucey/research>). Digital transformation is constructed using Sentence-BERT semantic classification and LDA topic model (*Digital*). We further categorize firm digital transformation into four key dimensions: Data (*Data*), Computing Power (*C-Power*), Algorithms (*Algorithm*), and the Integration of Informatization and Industrialization (*I-Industry*).² This classification provides a clearer framework for identifying firms' primary strategic focuses during the digital transformation process.

We also control for firm size (*Size*), the leverage ratio (*Lev*), return-on-assets (*ROA*), the return on operating cash flow (*Cashflow*), the fixed asset ratio (*FIXED*), the growth rate of operating revenue (*Growth*), board member (*Board*), the book-to-market ratio (*BM*), *Tobin Q*, the ownership structure (state-owned-enterprise (*SOE*)), and the listing age (*Age*), the institutional ownership ratio (*Institution*), the management ownership ratio (*Mshare*). The measurements of the control variables are provided in Table A.1. Table 1 presents the descriptive statistics for the core explanatory variable, dependent variable, and other control variables.

3.2. Empirical design

We construct the following model to examine the relationship between biodiversity risk and firm digital transformation:

$$Digital_{i,t} = \beta_0 + \beta_1 * Bio - Risk_{i,t} + \beta * \sum Controls_{i,t} + Industry FE + Year FE + \varepsilon_{i,t} \quad (1)$$

Where i and t represent the firm and the year, respectively. *Digital* is the dependent variable and represents firm digital transformation. *Bio-risk* is the independent variable and represents the firm's biodiversity risk exposure. *Controls* refer to a set of control variables. We account for industry and year-fixed effects.

4. Empirical analysis

4.1. Baseline results

Table 2 reports the baseline regression results, showing a significantly positive association between *Bio-Risk* and *Digital* at the 1 % level (coef. = 8.567, t-stat = 3.00), thus supporting H1. This result confirms that firms exposed to higher biodiversity risk are more likely to pursue digital transformation as a strategic response. Decomposing digital transformation into Data (*Data*), Computing Power (*C-Power*), Algorithms (*Algorithm*), and the Integration of Informatization and Industrialization (*I-Industry*) reveals heterogeneous corporate responses to biodiversity risk. Empirical results highlight a strategic prioritization sequence: *Data* > *I-Industry* > *C-Power* > *Algorithm*. Initially, firms prioritize building environmental data infrastructure to facilitate real-time monitoring, compliance, and supply chain risk assessment (Cong et al., 2024). Subsequently, firms integrate digital technologies with core operations to enhance resilience and resource efficiency (Gaspar et al., 2024). Investments in computing power, despite their importance for environmental analytics, typically occur later due to higher costs. Finally, algorithmic solutions, including AI-based simulations and predictive analytics, are implemented contingent upon established infrastructure and skilled personnel, positioning them as the concluding stage. Overall, firms adopt a progressive digital adaptation pathway from visibility and operational stability towards advanced analytics and automation to strategically manage biodiversity challenges.

² The detailed methodology for constructing the digital transformation indicator is provided in the Appendix Section A.1. The four dimensions systematically delineate distinct pathways for corporate digital transformation. Our analysis reveals that heterogeneity across industries and firm characteristics leads to differentiated strategic priorities in digital transformation dimensions in the Appendix Section A.1.

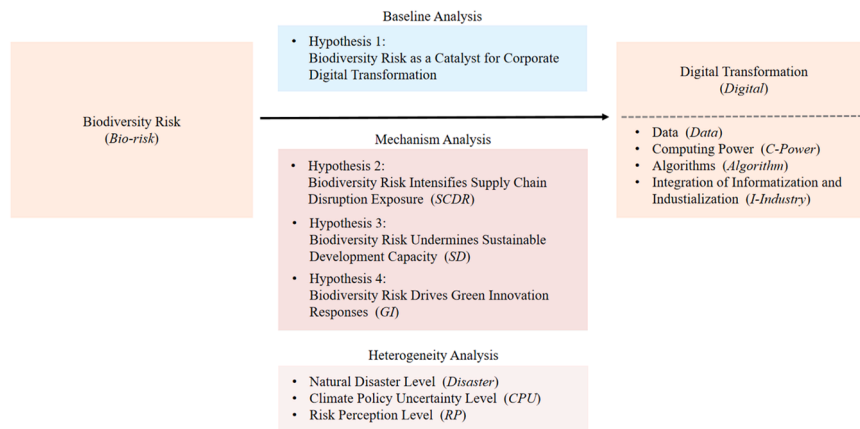


Fig. 1. Analytical framework.

Notes: This figure outlines the analytical framework.

Table 1
Summary statistics.

Variables	N	Mean	Median	SD	P25	P75
<i>Bio-Risk</i>	33,983	0.0001	0.0000	0.0003	0.0000	0.0001
<i>Digital</i>	33,983	0.0376	0.0095	0.1527	0.0014	0.0341
<i>Data</i>	33,983	0.0265	0.0052	0.1322	0.0006	0.0242
<i>C-Power</i>	33,983	0.0107	0.0015	0.0440	0.0002	0.0083
<i>Algorithm</i>	33,983	0.0102	0.0022	0.0404	0.0006	0.0076
<i>I-Industry</i>	33,983	0.0249	0.0042	0.0797	0.0005	0.0213
<i>SCDR</i>	33,983	−1.4725	−1.3002	1.0553	−1.9572	−0.7867
<i>SD</i>	33,983	3.9663	3.8618	0.5961	3.4324	4.4496
<i>GI</i>	33,983	0.2607	0.0000	0.6686	0.0000	0.0000
<i>Disaster</i>	33,983	9.0184	9.0526	0.4822	8.6109	9.3361
<i>CPU^a</i>	30,652	1.7411	1.7540	0.7327	1.2908	2.1926
<i>RP</i>	33,983	−0.0716	−0.0753	0.1492	−0.1727	0.0233
<i>Size</i>	33,983	22.2407	22.0348	1.2858	21.3136	22.9638
<i>Lev</i>	33,983	0.4076	0.3975	0.2024	0.2440	0.5551
<i>ROA</i>	33,983	0.0426	0.0403	0.0569	0.0144	0.0737
<i>Cashflow</i>	33,983	0.0491	0.0485	0.0650	0.0114	0.0888
<i>FIXED</i>	33,983	0.2046	0.1738	0.1531	0.0851	0.2902
<i>Growth</i>	33,983	0.1518	0.0922	0.4178	−0.0355	0.2436
<i>Board</i>	33,983	2.1111	2.1972	0.1952	1.9459	2.1972
<i>BM</i>	33,983	1.0248	0.6459	1.2031	0.3780	1.1555
<i>Tobin Q</i>	33,983	2.0185	1.6081	1.2998	1.2434	2.2918
<i>SOE</i>	33,983	0.3122	0.0000	0.4634	0.0000	1.0000
<i>Age</i>	33,983	2.0484	2.1972	0.9295	1.3863	2.8332
<i>Institution</i>	33,983	0.3745	0.3774	0.2378	0.1656	0.5614
<i>Mshare</i>	33,983	0.1495	0.0190	0.2017	0.0000	0.2831

Notes: This table reports the descriptive statistics of the variables. All variables are defined in the Appendix Table A.1.

^a Note that the Climate Policy Uncertainty (CPU) index is unavailable for firms registered in certain locations, including the Hong Kong Special Administrative Region, the Cayman Islands, the Ili Kazakh Autonomous Prefecture, the Horqin Gol City, the Xilin Gol League, the Alxa League, the Qiannan Buyei and Miao Autonomous Prefecture, the Wenshan Zhuang and Miao Autonomous Prefecture, and the Bayingolin Mongol Autonomous Prefecture.

To ensure the robustness of our results, we conducted endogeneity tests and heterogeneity analyses. The detailed results are provided in the Appendix Section A.2 and Section A.3, respectively.

4.2. Mechanism analysis

This section examines how biodiversity risk drives digital transformation, focusing on supply chain disruption risk, sustainable development, and green innovation. To test H2-H4, we include supply chain disruption risk (*SCDR*), sustainable development (*SD*), and green innovation (*GI*) as dependent variables. The corresponding empirical results are reported in Table 3.

Columns (1), (4), and (7) present the empirical results for the supply chain disruption risk mechanism. In column (1), the coefficient for *Bio-Risk* is significantly positive at the 1 % level, indicating that biodiversity risk substantially increases the likelihood of supply

Table 2
Biodiversity risk and digital transformation.

	Digital (1)	Data (2)	C-Power (3)	Algorithm (4)	I-Industry (5)
<i>Bio-Risk</i>	8.567*** (3.00)	7.463*** (2.95)	2.434*** (3.58)	2.194*** (3.57)	3.060** (2.31)
<i>Size</i>	0.006*** (4.08)	0.005*** (3.46)	0.002*** (4.99)	0.002*** (5.05)	0.003*** (4.79)
<i>Lev</i>	−0.002 (−0.39)	−0.004 (−0.93)	0.000 (0.21)	0.001 (0.42)	0.003 (0.91)
<i>ROA</i>	−0.046** (−2.01)	−0.030 (−1.51)	−0.015** (−2.17)	−0.017*** (−2.58)	−0.033*** (−2.88)
<i>Cashflow</i>	−0.044*** (−2.81)	−0.036*** (−2.67)	−0.010** (−2.35)	−0.007* (−1.86)	−0.022*** (−2.74)
<i>FIXED</i>	−0.034*** (−6.02)	−0.025*** (−5.37)	−0.010*** (−5.57)	−0.010*** (−5.45)	−0.020*** (−6.24)
<i>Growth</i>	−0.001 (−0.78)	−0.001 (−1.11)	0.000 (0.71)	0.000 (1.21)	−0.000 (−0.43)
<i>Board</i>	0.001 (0.32)	0.002 (0.45)	0.000 (0.27)	−0.000 (−0.11)	−0.000 (−0.03)
<i>BM</i>	−0.006*** (−7.21)	−0.004*** (−5.68)	−0.002*** (−7.95)	−0.002*** (−8.23)	−0.004*** (−9.87)
<i>Tobin Q</i>	−0.001 (−0.82)	−0.000 (−0.53)	−0.000 (−1.09)	−0.000 (−0.12)	−0.001* (−1.75)
<i>SOE</i>	−0.000 (−0.05)	0.000 (0.06)	0.000 (0.13)	0.001 (1.56)	−0.001 (−0.84)
<i>Age</i>	−0.003** (−2.20)	−0.002 (−1.51)	−0.001* (−1.86)	−0.001*** (−2.65)	−0.003*** (−3.89)
<i>Institution</i>	−0.011*** (−2.70)	−0.009** (−2.51)	−0.004*** (−3.41)	−0.004*** (−3.61)	−0.005** (−2.15)
<i>Mshare</i>	−0.012** (−2.02)	−0.008* (−1.68)	−0.006*** (−3.29)	−0.005*** (−3.64)	−0.006* (−1.77)
<i>Constant</i>	−0.066** (−2.18)	−0.058** (−2.07)	−0.021*** (−2.78)	−0.017** (−2.55)	−0.023* (−1.72)
<i>Year & Industry FE</i>	YES	YES	YES	YES	YES
<i>R²</i>	0.105	0.078	0.122	0.117	0.131
<i>N</i>	33,983	33,983	33,983	33,983	33,983

Notes: This table reports panel regression results examining the relationship between biodiversity risk and corporate digital transformation. The key independent variable is biodiversity risk exposure (*Bio-Risk*) for firm *i* in year *t*. The dependent variables include the overall level of digital transformation (*Digital*) and its four distinct dimensions for firm *i* in year *t*: Data (*Data*), Computing Power (*C-Power*), Algorithms (*Algorithm*), and Integration of Informatization and Industrialization (*I-Industry*). All regressions control for firm-level covariates, as well as year and industry fixed effects. Robust standard errors are clustered at the firm level. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively. T-statistics are reported in parentheses.

chain disruptions. Column (4) shows a significantly positive relationship between *SCDR* and digital transformation, suggesting that heightened supply chain risk serves as a key driver of digital adoption. In column (7), when both *Bio-Risk* and *SCDR* are included in the regression, the coefficient for *Bio-Risk* decreases from 8.567 (as reported in Table 2, column (1)) to 8.267, implying a reduction in explanatory power. Furthermore, both the Sobel and Goodman mediation tests confirm that supply chain disruption risk plays a significant mediating role in the relationship between biodiversity risk and digital transformation. Taken together, these results provide strong empirical support for H2.

Columns (2), (5), and (8), present the empirical results for the sustainable development mechanism. Column (2) shows that the coefficient of *Bio-Risk* is significantly negative at the 5 % level, indicating that biodiversity risk significantly constrains firms' sustainable development capacity. Column (5) reveals a significantly positive relationship between sustainable development (*SD*) and digital transformation, suggesting that firms with a stronger sustainability orientation are more likely to adopt digital strategies. In column (8), when both *Bio-Risk* and *SD* are included in the model, the coefficient of *Bio-Risk* increases from 8.567 (as reported in Table 2, column (1)) to 8.801, indicating an amplification of its effect. Furthermore, both the Sobel and Goodman tests confirm the mediating role of sustainable development in the pathway through which biodiversity risk influences digital transformation. Collectively, these findings provide strong support for H3.

Columns (3), (6), and (9) present the empirical results for the green innovation mechanism. Column (3) reports a significantly positive coefficient for *Bio-Risk* at the 10 % level, indicating that biodiversity risk significantly promotes firms' engagement in green innovation. Column (6) further reveals that green innovation (*GI*) exerts a significant positive effect on digital transformation, suggesting that ecological pressures drive firms to pursue digital solutions through environmental innovation. When both *Bio-Risk* and *GI* are included in the regression in Column (9), the coefficient of *Bio-Risk* decreases from 8.567 (Table 2, Column 1) to 8.333, reflecting a partial mediation effect. In addition, both Sobel and Goodman mediation tests confirm the significance of green innovation as a mediating channel through which biodiversity risk affects digital transformation. Taken together, these results provide robust support for H4.

Table 3

Supply chain disruption risk channel, sustainable development channel, and green innovation channel.

	<i>SCDR</i>	<i>SD</i>	<i>GI</i>	<i>Digital</i>					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>Bio-Risk</i>	0.786*** (2.65)	−48.376** (−2.13)	0.361* (1.82)				8.267*** (2.93)	8.801*** (3.11)	8.333*** (2.97)
<i>SCDR</i>				0.391*** (3.85)			0.381*** (3.77)		
<i>SD</i>					0.005** (2.55)			0.005*** (2.61)	
<i>GI</i>						0.653** (2.47)			0.646** (2.46)
<i>Sobel</i>							0.619*** (4.70)	−0.079* (−1.68)	0.992*** (5.75)
<i>Goodman-1</i>							0.619*** (4.67)	−0.079 (−1.62)	0.992*** (5.74)
<i>Goodman-2</i>							0.619*** (4.72)	−0.079* (−1.75)	0.992*** (5.77)
<i>Constant</i>	0.002 (0.53)	3.125*** (13.49)	−0.036*** (−9.51)	−0.070** (−2.29)	−0.085*** (−2.71)	−0.046 (−1.62)	−0.067** (−2.20)	−0.082*** (−2.64)	−0.043 (−1.53)
<i>Controls</i>	YES	YES	YES	YES	YES	YES	YES	YES	YES
<i>Year & Industry FE</i>	YES	YES	YES	YES	YES	YES	YES	YES	YES
<i>R²</i>	0.427	0.130	0.177	0.105	0.105	0.105	0.105	0.105	0.106
<i>N</i>	33,983	33,983	33,983	33,983	33,983	33,983	33,983	33,983	33,983

Notes: This table presents mediation analyses exploring how biodiversity risk influences firms' digital transformation through three channels: supply chain disruption risk, sustainable development, and green innovation. The key independent variable is the biodiversity risk (*Bio-Risk*) for firm *i* in year *t*. The dependent variables are supply chain disruption risk (*SCDR*), sustainable development (*SD*), and green innovation (*GI*) for firm *i* in year *t*, respectively. Columns (1), (4), and (7) report results for the supply chain disruption channel, proxied by the natural logarithm of the ratio of accounts receivable and prepayments to main business revenue. Columns (2), (5), and (8) capture the sustainable development channel, measured by the natural logarithm of the inverse of energy consumption per million people. Columns (3), (6), and (9) reflect the green innovation channel, proxied by the natural logarithm of the number of independently filed green patents. All regressions control for firm-level covariates, as well as year and industry fixed effects. Robust standard errors are clustered at the firm level. ***, **, and * denote statistical significance at the 1 %, 5 %, and 10 % levels, respectively. T-statistics (or Z-values from mediation tests, including Sobel and Goodman) are reported in parentheses.

Furthermore, Table 4, Table 5, Table 6 present a detailed examination of the mediating mechanisms through which biodiversity risk affects the multi-dimensional pathways of firms' digital transformation. Table 4 indicates that supply chain disruption risk (*SCDR*) significantly mediates this relationship across all four transformation dimensions—Data, Computing Power, Algorithms, and the Integration of Informatization and Industrialization—evidenced by consistently positive and significant coefficients. The inclusion of *SCDR* reduces the coefficient of *Bio-Risk* compared to Table 2, suggesting partial mediation. Sobel and Goodman tests further confirm this effect. Table 5 investigates sustainable development (*SD*) as a mediator, showing its positive influence on all dimensions. Notably, controlling for *SD* increases the coefficient of *Bio-Risk*, indicating that *SD* strengthens its direct effect. This mediating role is significant across all dimensions except for the Integration of Informatization and Industrialization. Table 6 highlights green innovation (*GI*) as another mechanism. *GI* consistently promotes digital transformation, and its inclusion weakens the effect of *Bio-Risk*, supporting partial mediation. Again, the Sobel and Goodman tests validate the statistical significance of this pathway. Collectively, the results demonstrate that *SCDR*, *SD*, and *GI* serve as critical channels through which biodiversity risk drives digital transformation.

5. Conclusion

This study investigates whether biodiversity risk accelerates firms' digital transformation. Our results confirm a significant positive effect, especially for firms facing greater natural disaster risks, climate policy uncertainty, and heightened risk perceptions. Digital transformation is analyzed across four dimensions—Data, Integration of Informatization and Industrialization, Computing Power, and Algorithms—in order of firms' emphasis. The primary drivers are increased supply chain disruption, limited sustainable development opportunities, and greater incentives for green innovation.

These findings offer several important practical implications. From a business strategy perspective, firms operating in high-biodiversity-risk environments should prioritize digital transformation as a resilience-enhancing tool. Specifically, adopting advanced data analytics, improving algorithmic forecasting, and strengthening information technology–operational technology integration can help firms better monitor environmental risks, anticipate disruptions, and optimize decision-making. For policy-makers, governments should consider providing targeted fiscal incentives (e.g., tax credits, subsidies) for green digital investments, especially in sectors highly dependent on ecosystem services. Additionally, enhancing digital infrastructure in ecologically sensitive regions and promoting cross-sectoral platforms for environmental risk data sharing can accelerate the digital transition. Regulatory bodies can also encourage biodiversity risk disclosure and offer standardized guidelines to support corporate risk adaptation through digital means.

Table 4

Supply chain disruption risk and the multi-dimensional pathways of digital transformation.

	<i>Data</i>		<i>C-Power</i>		<i>Algorithm</i>		<i>I-Industry</i>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>SCDR</i>	0.279*** (3.05)	0.271*** (2.98)	0.125*** (4.61)	0.123*** (4.52)	0.130*** (5.37)	0.127*** (5.28)	0.233*** (4.86)	0.229*** (4.81)
<i>Bio-Risk</i>		7.250*** (2.90)		2.338*** (3.48)		2.094*** (3.45)		2.880** (2.20)
<i>Sobel</i>		0.420*** (4.28)		0.196*** (4.83)		0.183*** (4.85)		0.422*** (5.04)
<i>Goodman-1</i>		0.420*** (4.25)		0.196*** (4.81)		0.183*** (4.83)		0.422*** (5.03)
<i>Goodman-2</i>		0.420*** (4.31)		0.196*** (4.85)		0.183*** (4.87)		0.422*** (5.06)
<i>Constant</i>	−0.061** (−2.16)	−0.058** (−2.08)	−0.022*** (−2.91)	−0.021*** (−2.80)	−0.018*** (−2.69)	−0.018*** (−2.58)	−0.024* (−1.83)	−0.023* (−1.75)
<i>Controls</i>	YES	YES	YES	YES	YES	YES	YES	YES
<i>Year & Industry FE</i>	YES	YES	YES	YES	YES	YES	YES	YES
<i>R²</i>	0.078	0.079	0.122	0.123	0.118	0.118	0.131	0.131
<i>N</i>	33,983	33,983	33,983	33,983	33,983	33,983	33,983	33,983

Notes: This table presents the mediation analysis of how biodiversity risk influences firms' multi-dimensional digital transformation through the supply chain disruption risk channel. The key independent variables are biodiversity risk (*Bio-Risk*) and supply chain disruption risk (*SCDR*) for firm *i* in year *t*. The dependent variables are multi-dimensional digital transformation for firm *i* in year *t*: Data (*Data*), Computing Power (*C-Power*), Algorithms (*Algorithm*), and the Integration of Informatization and Industrialization (*I-Industry*). All regressions control for firm-level covariates, as well as year and industry fixed effects. Robust standard errors are clustered at the firm level. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively. T-statistics (or Z-values from mediation tests, including Sobel and Goodman) are reported in parentheses.

Table 5

Sustainable development and the multi-dimensional pathways of digital transformation.

	<i>Data</i>		<i>C-Power</i>		<i>Algorithm</i>		<i>I-Industry</i>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>SD</i>	0.004*** (2.79)	0.005*** (2.86)	0.001** (2.55)	0.001*** (2.61)	0.001** (2.47)	0.001** (2.53)	0.001 (1.05)	0.001 (1.09)
<i>Bio-Risk</i>		7.686*** (3.06)		2.502*** (3.70)		2.253*** (3.68)		3.109** (2.35)
<i>Sobel</i>		−0.084* (−1.78)		−0.030* (−1.81)		−0.025* (−1.76)		0.006 (0.45)
<i>Goodman-1</i>		−0.084* (−1.73)		−0.030* (−1.77)		−0.025* (−1.71)		0.006 (0.41)
<i>Goodman-2</i>		−0.084* (−1.83)		−0.030* (−1.86)		−0.025* (−1.82)		0.006 (0.52)
<i>Constant</i>	−0.075*** (−2.60)	−0.072** (−2.54)	−0.026*** (−3.37)	−0.026*** (−3.28)	−0.022*** (−3.09)	−0.021*** (−3.00)	−0.027** (−2.01)	−0.026* (−1.93)
<i>Controls</i>	YES	YES	YES	YES	YES	YES	YES	YES
<i>Year & Industry FE</i>	YES	YES	YES	YES	YES	YES	YES	YES
<i>R²</i>	0.078	0.079	0.122	0.122	0.117	0.117	0.131	0.131
<i>N</i>	33,983	33,983	33,983	33,983	33,983	33,983	33,983	33,983

Notes: This table presents the mediation analysis of how biodiversity risk influences firms' multi-dimensional digital transformation through the sustainable development channel. The key independent variables are biodiversity risk (*Bio-Risk*) and sustainable development (*SD*) for firm *i* in year *t*. The dependent variables are multi-dimensional digital transformation for firm *i* in year *t*: Data (*Data*), Computing Power (*C-Power*), Algorithms (*Algorithm*), and the Integration of Informatization and Industrialization (*I-Industry*). All regressions control for firm-level covariates, as well as year and industry fixed effects. Robust standard errors are clustered at the firm level. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively. T-statistics (or Z-values from mediation tests, including Sobel and Goodman) are reported in parentheses.

CRedit authorship contribution statement

Libo Yin: Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Xiaoye Zhu:** Writing – review & editing, Writing – original draft, Software, Methodology, Data curation. **Jingtian Li:** Writing – original draft, Software, Data curation.

Declaration of competing interest

I, Libo Yin, declare that there are no conflicts of interest in relation to the manuscript titled “Does exposure to biodiversity risk drive firms' digital transformation?” submitted to *Finance Research Letters*. I confirm that the results and interpretations reported in the

Table 6
Green Innovation and the multi-dimensional pathways of digital transformation.

	<i>Data</i>		<i>C-Power</i>		<i>Algorithm</i>		<i>I-Industry</i>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>GI</i>	0.469*	0.463*	0.173***	0.170***	0.159***	0.157***	0.431***	0.429***
	(1.90)	(1.88)	(2.70)	(2.68)	(2.92)	(2.89)	(4.01)	(4.00)
<i>Bio-Risk</i>		7.296***		2.373***		2.137***		2.905**
		(2.95)		(3.54)		(3.53)		(2.21)
<i>Sobel</i>		0.712***		0.282***		0.251***		0.642***
		(5.44)		(5.74)		(5.69)		(6.03)
<i>Goodman-1</i>		0.712***		0.282***		0.251***		0.642***
		(5.42)		(5.72)		(5.67)		(6.02)
<i>Goodman-2</i>		0.712***		0.282***		0.251***		0.642***
		(5.46)		(5.75)		(5.70)		(6.05)
<i>Constant</i>	−0.043*	−0.041	−0.016**	−0.015*	−0.012*	−0.012*	−0.008	−0.007
	(−1.71)	(−1.62)	(−2.05)	(−1.95)	(−1.80)	(−1.69)	(−0.63)	(−0.55)
<i>Controls</i>	YES	YES	YES	YES	YES	YES	YES	YES
<i>Year & Industry FE</i>	YES	YES	YES	YES	YES	YES	YES	YES
<i>R²</i>	0.079	0.079	0.122	0.123	0.117	0.118	0.132	0.132
<i>N</i>	33,983	33,983	33,983	33,983	33,983	33,983	33,983	33,983

Notes: This table presents the mediation analysis of how biodiversity risk influences firms' multi-dimensional digital transformation through the green innovation channel. The key independent variables are biodiversity risk (*Bio-Risk*) and green innovation (*GI*) for firm *i* in year *t*. The dependent variables are multi-dimensional digital transformation for firm *i* in year *t*: *Data* (*Data*), Computing Power (*C-Power*), Algorithms (*Algorithm*), and the Integration of Informatization and Industrialization (*I-Industry*). All regressions control for firm-level covariates, as well as year and industry fixed effects. Robust standard errors are clustered at the firm level. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively. T-statistics (or Z-values from mediation tests, including Sobel and Goodman) are reported in parentheses.

manuscript are original and have not been plagiarized.

I certify that I have read and understand the *Finance Research Letters* conflict of interest policy, and I understand that failure to disclose a conflict of interest may result in the manuscript being rejected or retracted.

I also certify that I have disclosed any financial or non-financial relationships that may be interpreted as constituting a conflict of interest in relation to this manuscript. I understand that this information will be subject to peer review, and I am willing to provide further information or clarification if required.

I confirm that I have no known conflicts of interest that would influence the results or interpretation of the data presented in this manuscript, and I understand that failure to disclose a conflict of interest is unethical and may result in sanctions being imposed on me.

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Appendix

A.1. Construction of the firms' digital transformation

We employ a combination of Sentence-BERT (a variant of the Bidirectional Encoder Representations from Transformers (BERT) model) semantic classification and LDA topic model as a text analysis approach to derive categorized digitalization information, and then apply machine-learning dimensionality-reduction techniques to construct a composite digitalization index.

We leverage the strengths of Sentence-BERT in classifying digitalization-related statements, alongside the unsupervised modeling capacity of the LDA model to extract latent digitalization clusters from textual data. This integrated approach enables automated identification of key topics related to corporate digital transformation from vast textual sources (Bellstam et al., 2021; Bybee et al., 2023). To align these diverse digitalization dimensions into a unified indicator framework, thus reducing data complexity while retaining core features needed for comparative analysis across firms, we apply dimensionality reduction through Sparse Principal Component Analysis (Sparse PCA). This method simplifies the data structure through linear transformations, extracts the most representative digital features, and introduces sparsity in feature selection. By applying regularization, we ensure that only the most significant features remain in the principal components, effectively eliminating irrelevant or redundant information. This process aims to enhance the simplicity of the digitalization indicators and improve the efficiency of information interpretation.

The process of constructing corporate digitalization indicators involves several key steps. First, we use national digital-economy policy documents, industry white papers on digital technologies, and annual reports of A-share listed companies in China as textual sources (2007–2023), all texts are pre-processed with Chinese word segmentation and stop word removal. Second, we use Sentence-BERT to perform semantic classification and embedding transformation to capture the deep semantic information related to digitalization. Third, we determine the optimal number of topics through K-Means clustering and extract core topics from the text using the

LDA model. Cosine-similarity matching between firm-level texts and these themes yields four dimension-specific digitalisation scores. Finally, we apply machine learning-based dimensionality-reduction methods to capture differentiated features and condense these scores into a single composite index of corporate digitalisation. The process flow is shown in Fig. A.1.

Table A.1

Description of the regression variables.

Variables	Description
<i>Bio-Risk</i>	The proportion of characters related to biodiversity in the annual report to the total number of characters in the report
<i>Digital</i>	This indicator measures the degree of corporate digitalization and is constructed based on topics extracted from corporate annual reports using an LDA model. The specific methodology for its construction is detailed in the appendix
<i>Data</i>	This indicator quantifies the degree to which a company prioritizes the “Data” dimension or strategy in its digital transformation initiatives
<i>C-Power</i>	This indicator quantifies the extent to which a company prioritizes the “Computing Power” dimension or strategy in its digital transformation efforts
<i>Algorithm</i>	This indicator quantifies the extent to which a company prioritizes the “Algorithm” dimension or strategy in its digital transformation initiatives
<i>I-Industry</i>	This indicator quantifies the extent to which a company prioritizes the “Integration of Informatization and Industrialization” dimension or strategy in its digital transformation initiatives
<i>SCDR</i>	The natural logarithm of the ratio of accounts receivable and prepayments to main business revenue
<i>SD</i>	The natural logarithm of the inverse of energy consumption per million people
<i>GI</i>	The natural logarithm of the number of independently filed green patents
<i>Disaster</i>	The natural logarithm of the frequency of natural disasters
<i>CPU</i>	Climate policy uncertainty index developed by Ma et al. (2023)
<i>RP</i>	The annual skewness of daily stock returns
<i>Size</i>	The natural logarithm of total assets at year-end
<i>Lev</i>	The ratio of total liabilities to total assets
<i>ROA</i>	The ratio of net income to total assets
<i>Cashflow</i>	The ratio of the net cash flow from operating activities to total assets
<i>FIXED</i>	The ratio of net fixed assets to total assets
<i>Growth</i>	The growth rate of operating revenue
<i>Board</i>	The natural logarithm of the number of board members
<i>BM</i>	The ratio of book value to total market value
<i>Tobin Q</i>	The sum of total assets plus market value of equity minus book value of equity divided by total assets
<i>SOE</i>	An indicator variable equals 1 if the firm is a state-owned enterprise, and 0 otherwise.
<i>Age</i>	The natural logarithm of years listed
<i>Institution</i>	The total number of shares held by institutional investors divided by the total outstanding shares
<i>Mshare</i>	The management ownership divided by total shares outstanding

(1) Overview of the construction of categorized digitalization indicators across four dimensions

Below, we formally outline the system of equations that underpins the construction of the study’s digitalisation indicators. The core of Sentence-BERT lies in converting an input sentence $S = \{s_1, s_2, \dots, s_n\}$ into a fixed-dimensional vector representation v . This process can be expressed as:

$$v = SBERT(S) \quad (A1)$$

In this formulation, Sentence-BERT utilizes a multi-layer bidirectional Transformer encoder to map each word s_i in the sentence to a high-dimensional vector space, and then performs pooling to generate the overall sentence embedding vector. Specifically, given the input sentence S , the embedding vector is computed as:

$$v = Pool(Transformer(S)) \quad (A2)$$

In semantic classification tasks, Sentence-BERT is trained such that sentences with similar meanings are closer in the vector space, thereby enhancing classification accuracy and efficiency.

Next, the semantically classified sentences are input into the LDA model. The core idea of LDA is to assign a topic distribution to each document and a word distribution to each topic. Specifically, for a given document d with vocabulary $w = \{w_1, w_2, \dots, w_N\}$, the LDA model generates data through the following steps:

- a) For each document d , a topic distribution θ_d is drawn from a Dirichlet distribution:

$$\theta_d \sim Dir(\alpha) \quad (A3)$$

- b) For each word w_i :

1. A topic z_i is drawn from the topic distribution θ_d :

$$z_i \sim \text{Multinomial}(\theta_d) \quad (\text{A4})$$

2. A word w_i is drawn from the word distribution β_{z_i} :

$$w_i \sim \text{Multinomial}(\beta_{z_i}) \quad (\text{A5})$$

The objective of LDA topic identification is to estimate the topic distribution θ and word distribution β by maximizing the likelihood function:

$$L(\theta, \beta | w) = \prod_{d=1}^D \prod_{n=1}^{N_d} \sum_{z=1}^K p(z | \theta_d) p(w_{dn} | \beta_z) \quad (\text{A6})$$

where D is the total number of documents, N_d is the number of words in document d , and K is the number of topics.

Through methods such as variational inference or Gibbs sampling, the LDA model can effectively extract representative topics from large volumes of text. This, in turn, aids in optimizing the construction and analysis of categorized digitalization indicators. Building on the analytical framework, our methodology yields four dimensions of firm-level digitalization indicators: Data, Computing Power, Algorithms, and Integration of Informatization and Industrialization.³

(2) Overview of Constructing a Composite Digitalization Index Using Sparse PCA for Dimensionality Reduction

After obtaining the categorized digitalization indicators for each company, we applied machine learning-based Sparse PCA for dimensionality reduction, in order to consolidate the various digitalization dimensions into a unified indicator framework. The following is a brief explanation of this method and its main formulas.

Sparse PCA extends traditional PCA by introducing a sparsity constraint to improve the interpretability of the principal components. By limiting the number of non-zero loadings in each principal component, Sparse PCA ensures that each component is composed of a linear combination of only a few original variables, making it easier to understand and interpret the data structure. The key formulas for this method are as follows:

For a standardized data matrix $X \in \mathbb{R}^{n \times p}$, the covariance matrix is:

$$\Sigma = \frac{1}{n-1} X^T X \quad (\text{A7})$$

Through eigenvalue decomposition, we obtain:

$$\Sigma = Q \Lambda Q^T \quad (\text{A8})$$

where Q is the matrix of eigenvectors of the covariance matrix, and Λ is the diagonal matrix of eigenvalues. The i -th principal component PC_i can be expressed as:

$$PC_i = X q_i \quad (\text{A9})$$

where q_i is the i -th eigenvector.

Building on the above PCA method, Sparse PCA introduces a regularization term to enforce sparsity:

$$\max_w w^T \Sigma w - \lambda \|w\|_1 \quad (\text{A10})$$

where Σ is the covariance matrix, w is the principal component loading vector, and λ is the regularization parameter controlling the sparsity.

By applying Sparse PCA for dimensionality reduction, this study effectively captures both the global structure and local complexity of the original digitalization data from multiple perspectives. This not only reduces the dimensionality and complexity of the data, but also significantly enhances the accuracy and reliability of the quantitative evaluation, providing a solid foundation for comparative analysis of corporate digitalization levels. [Figs. A.2 and A.3](#) illustrate the variations in digitalization levels over time and across industries, respectively.

³ Structurally, the Data dimension captures organizational competencies in data management, cloud storage, and big data technologies. The Computing Power dimension reflects firm-level capabilities in cloud computing architectures, processor utilization, and information processing systems. The Algorithms dimension encompasses machine learning applications including quantitative modeling, performance optimization, and artificial intelligence implementations. The Integration of Informatization and Industrialization dimension evaluates firms' potential in industrial digitization, intelligent transformation, and digital-real economy convergence.

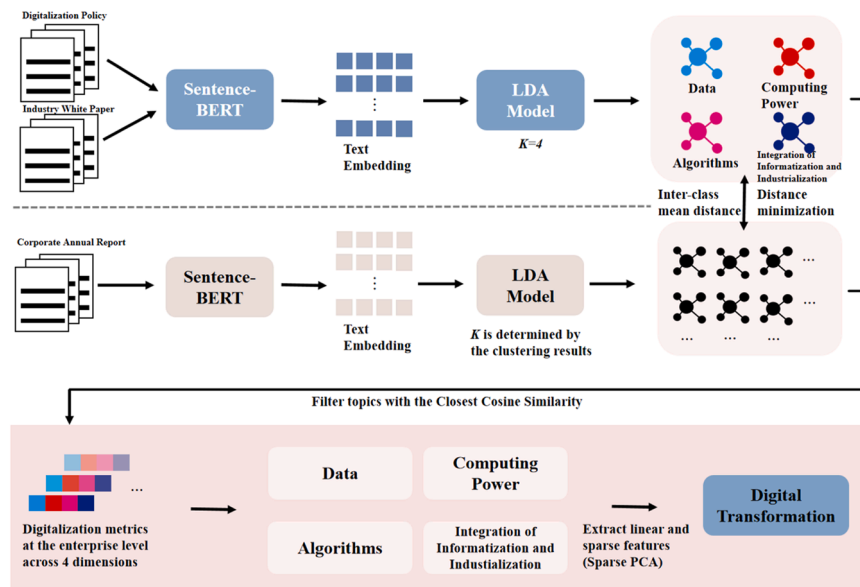


Fig. A.1. Firm Digital Transformation Index Processing Flowchart.

Notes: This figure outlines the methodology for constructing the digital transformation index.

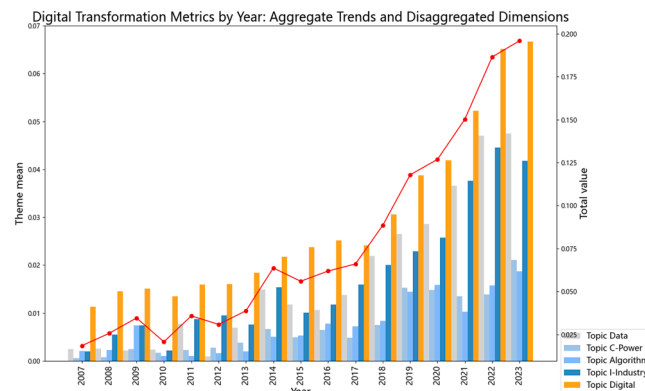


Fig. A.2. Annual trend of digitalization scores.

Notes: This figure illustrates the variations in digitalization levels over time.

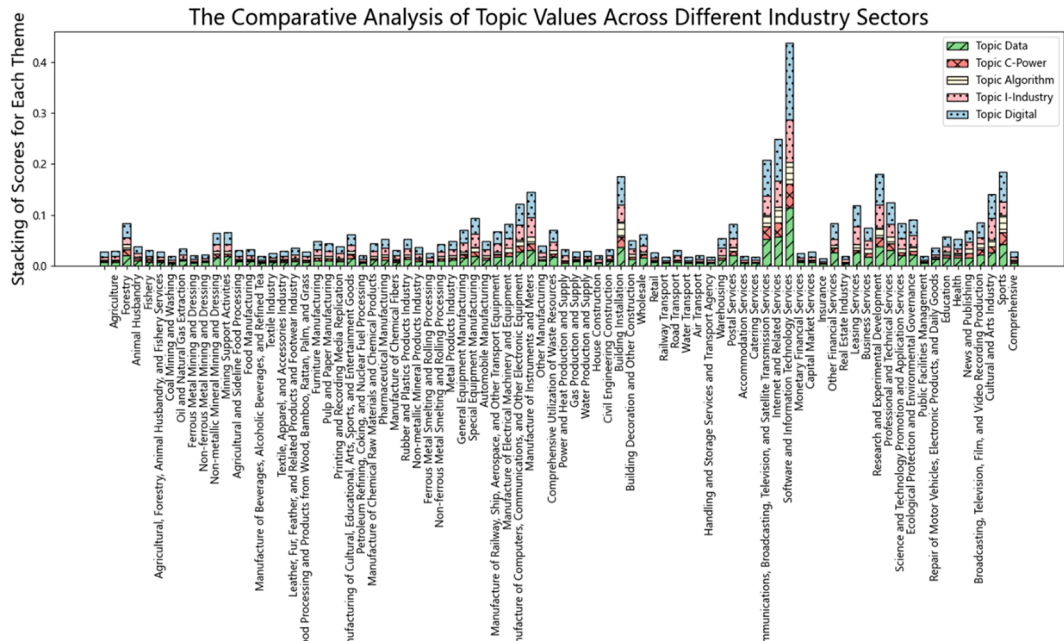


Fig. A.3. Industry heterogeneity of digitalization scores.
Notes: This figure illustrates the variations in digitalization levels across industries.

A.2. Endogeneity test

We use two methods to assess endogeneity: the propensity scores matching method (PSM) and two-stage least squares regression.

A.2.1. Propensity score matching method. We employ propensity score matching (PSM) to mitigate potential endogeneity issues. As biodiversity risk is a continuous variable, we transform it into a discrete binary variable. We assign a value of one to companies in the top one-third of biodiversity risk, representing the high biodiversity risk group (treatment group), and zero to companies in the bottom one-third, representing the low biodiversity risk group (control group). We select a series of variables that impact firm digital transformation as matching variables: firm size, the leverage ratio, the return-on-assets ratio, return on operating cash flow, the fixed asset ratio, growth rate of operating revenue, the number of board members, the book-to-market ratio, Tobin Q, ownership structure, listing age, the institutional ownership ratio, and the management ownership ratio. After matching, differences in the matching variables between the two groups are no longer significant. The empirical results are presented in Table A.2. The matching results indicate that the difference between the experimental group and the control group is 0.024 (with a t value = 11.31). This finding indicates that firms with greater biodiversity risk tend to leverage digital transformation to manage and mitigate such risks.

Table A.2
Endogeneity test: PSM method.

	Digital
Bio-Risk	0.024*** (11.31)
Constant	0.028*** (18.68)
Controls	YES
Year & Industry FE	YES
R ²	0.006
N	22,525

Notes: This table reports propensity score matching (PSM) to mitigate potential endogeneity issues. The dependent variable is the level of corporate digital transformation for the firm *i* in year *t* (*Digital*), and the independent variable is the biodiversity risk exposure for the firm *i* in year *t* (*Bio-Risk*). Control variables are included in all regressions, along with year and industry-fixed effects. Robust standard errors are

clustered at the firm level. ***, **, and * denote significance at the 1 %, 5 %, and 10 % levels, respectively. T-statistics are shown in parentheses.

A.2.2. Two-stage least-squares regression

We adopt the two-stage least-squares instrumental variable approach to address the endogeneity problem. We use the industry-year average biodiversity risk of other firms as the instrumental variable, denoted as *Ind-Bio-Risk*. The results of the weak instrumental variable test show that the minimum eigenvalue statistic is $1021.53 > 16.38$ (2SLS size of nominal 5 % Wald test), indicating that the instrumental variables are valid. The empirical results are presented in Table A.3. Column (1) reports the results of the first stage, which regresses *Bio-Risk* on the instrumental variables and other controls. The coefficient of the instrumental variable is significant at the 1 % level. Column (2) reports the results of the second stage, where the coefficient of *Bio-Risk* is 18.606 and is significantly negative at the 1 % level. Therefore, our previous finding is that firms' exposure to biodiversity risks significantly drives their digital transformation.

Table A.3
Endogeneity test: 2SLS regression.

	<i>Bio-Risk</i> First Stage (1)	<i>Digital</i> Second Stage (2)
<i>Ind-Bio-Risk</i>	1.000*** (7.20)	
<i>Bio-Risk</i>		18.606*** (3.13)
<i>Constant</i>	−0.000*** (−4.52)	−0.111*** (−3.66)
<i>Controls</i>	YES	YES
<i>Year & Industry FE</i>	YES	YES
<i>R</i> ²	0.212	0.104
<i>N</i>	33,983	33,983

Notes: This table reports the regression results of a two-stage least-squares instrumental variable approach to alleviating the endogeneity problem. The instrumental variable is the industry-year average biodiversity risk (*Ind-Bio-Risk*) of other firms. Column (1) reports the results of the first stage, which regresses the biodiversity risk (*Bio-Risk*) on the instrumental variables. Column (2) reports the results of the second stage, which regresses corporate digital transformation (*Digital*) on biodiversity risk (*Bio-Risk*). Control variables are included in all regressions, along with year and industry-fixed effects. Robust standard errors are clustered at the firm level. ***, **, and * denote significance at the 1 %, 5 %, and 10 % levels, respectively. T-statistics are shown in parentheses.

A.3. Heterogeneity analysis

Biodiversity risk and natural disaster-related challenges in the form of desertification, hurricanes, typhoons, droughts, coastal flooding, forest fires, and displacement pose significant economic risks to the firm. Severe natural disasters heighten firms' exposure to biodiversity risk, increasing the need for effective risk management tools. The results of this regression analysis, presented in Table A.4, suggest that the interaction between biodiversity risk and natural disaster frequency further exacerbates challenges to digital transformation. This effect is particularly evident in the significantly larger regression coefficient for *Bio-Risk* in the *High-Disaster* group compared to the *Low-Disaster* group.

Motivated by the assertion that biodiversity risk is distinct from climate-related risks, we investigate how the combination of these two risk factors impacts a firm digital transformation. The results of this regression analysis, presented in Table A.4, suggest that the interaction between biodiversity risk and climate policy uncertainty may exacerbate challenges to firm digital transformation. This is particularly evident in the significantly larger regression coefficient for *Bio-Risk* in the *High-CPU* group compared to the *Low-CPU* group.

Given that elevated risk perception enhances firms' sensitivity to external threats and motivates the adoption of advanced technologies to maintain operational stability, risk perception, and biodiversity risk may jointly drive digital transformation. Regression results presented in Table A.4 confirm that this interaction significantly intensifies the impact of biodiversity risk on digital transformation, as evidenced by a notably larger regression coefficient for *Bio-Risk* in the *High-RP* group than in the *Low-RP* group.

Table A.4
Heterogeneity analysis.

	<i>Digital</i>					
	<i>Low-Disaster</i> (1)	<i>High-Disaster</i> (2)	<i>Low-CPU</i> (3)	<i>High-CPU</i> (4)	<i>Low-RP</i> (5)	<i>High-RP</i> (6)
<i>Bio-Risk</i>	1.660 (1.43)	9.808** (2.38)	8.851** (2.06)	9.702*** (2.65)	5.826** (2.14)	12.526*** (2.81)
Constant	0.012 (1.05)	−0.141** (−2.34)	−0.081 (−1.58)	−0.061 (−1.59)	−0.090** (−2.04)	−0.034 (−0.96)
<i>Controls</i>	YES	YES	YES	YES	YES	YES
<i>Year & Industry FE</i>	YES	YES	YES	YES	YES	YES
R^2	0.140	0.106	0.094	0.115	0.115	0.101
N	16,988	16,995	15,321	15,329	16,988	16,995

Notes: This table presents grouped panel regressions of corporate digital transformation on biodiversity risk. The dependent variable is corporate digital transformation (*Digital*) for firm i in year t , the independent variable is biodiversity risk (*Bio-Risk*) for firm i in year t . Columns (1) to (2) represent the full sample divided into *Low-Disaster* and *High-Disaster* groups based on the value of the natural logarithm of the frequency of natural disasters. Columns (3) to (4) represent the full sample divided into *Low-CPU* and *High-CPU* groups based on the value of the climate policy uncertainty index developed by Ma et al. (2023) (<http://www.cnfn.com/data/download/climate-risk-database/>). Columns (5) to (6) represent the full sample divided into *Low-RP* and *High-RP* groups based on the value of the annual skewness of daily stock returns. Control variables are included in all regressions, along with year and industry-fixed effects. Robust standard errors are clustered at the firm level. ***, **, and * denote significance at the 1 %, 5 %, and 10 % levels, respectively. T-statistics are shown in parentheses.

Data availability

Data will be made available on request.

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